

Exp 1: PRINT A MESSAGE USING UART CONSOLE

**Name: Pradip Maity
CDSID: PMAITY**



Table of Contents

S.No.		Page No
1	Introduction	3
2	Scope	3
3	Prerequisite	3
4	Connection Diagram	3
5	Working Principle	4
6	Application Code	4
7	Output	5
8	Hardware Image	5
9	Appendix	5

1. Introduction:

In this project, we will explore how to print a simple message on a UART serial console using the Raspberry Pi Pico microcontroller. UART (Universal Asynchronous Receiver Transmitter) is a hardware communication protocol used for serial data transmission between devices. It allows data to be sent and received one bit at a time without requiring a clock signal.

The program will transmit a predefined message to a PC through UART communication, allowing us to view the output in real time using a terminal application such as PuTTY. This project helps in understanding basic serial communication and how microcontrollers interact with external systems.

2. Scope:

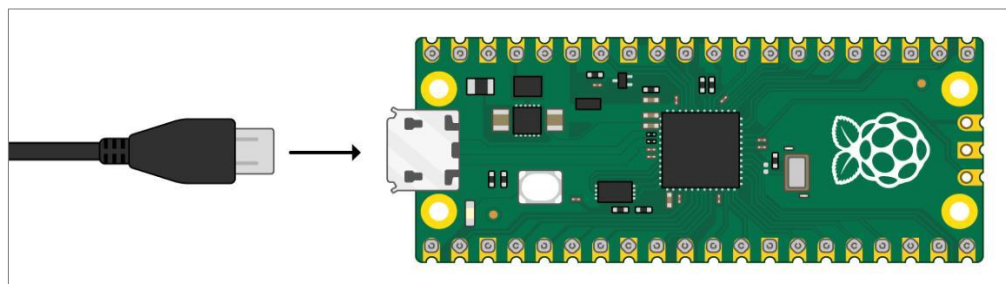
This project provides a foundational understanding of UART-based serial communication. It covers the setup of UART on the Raspberry Pi Pico and demonstrates how to send data to a serial console. The knowledge gained from this project is essential for debugging embedded systems, monitoring data, and enabling communication between devices in applications such as IoT and automation.

3. Prerequisite:

Hardware: Raspberry Pi Pico Microcontroller, USB Connector, Breadboard

Software: VS Code (with Pico SDK), PuTTY Serial Console (in Windows)

4. Connection Diagram:



5. Working Principle:

Printing a message in a UART serial console involves initializing UART communication on the Raspberry Pi Pico and transmitting data through it. The Pico is connected to the PC via a USB cable, which also provides a USB-to-serial interface. After powering the board, the system detects a virtual COM port on the PC.

The UART is initialized with a baud rate of 115200, and specific GPIO pins are configured for transmit (TX) and receive (RX) functions. Once initialized, the microcontroller sends a string message through UART. This message travels over the serial connection and is received by the PC.

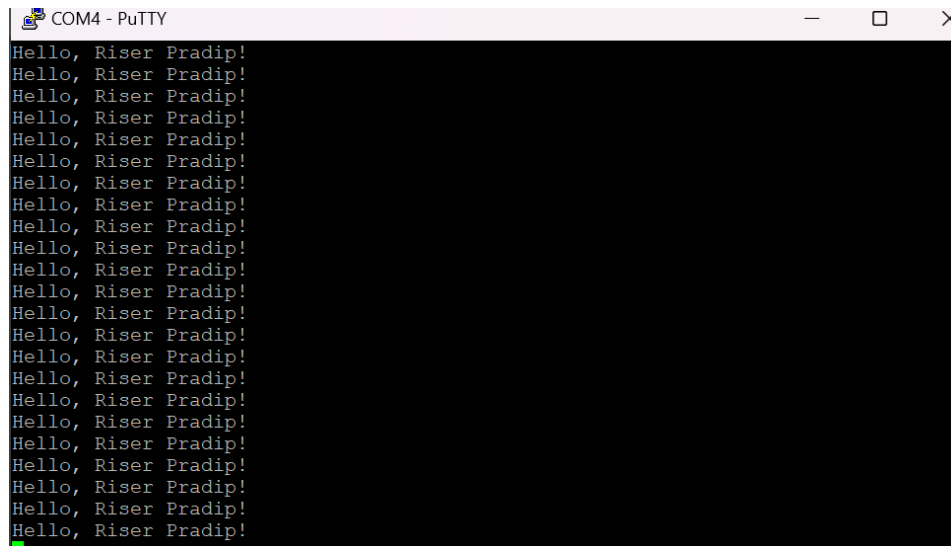
A terminal application like PuTTY is configured with the correct COM port and baud rate to display the incoming data. The program continuously sends the message at regular intervals, allowing the user to observe it in real time on the UART console.

6. Application code:



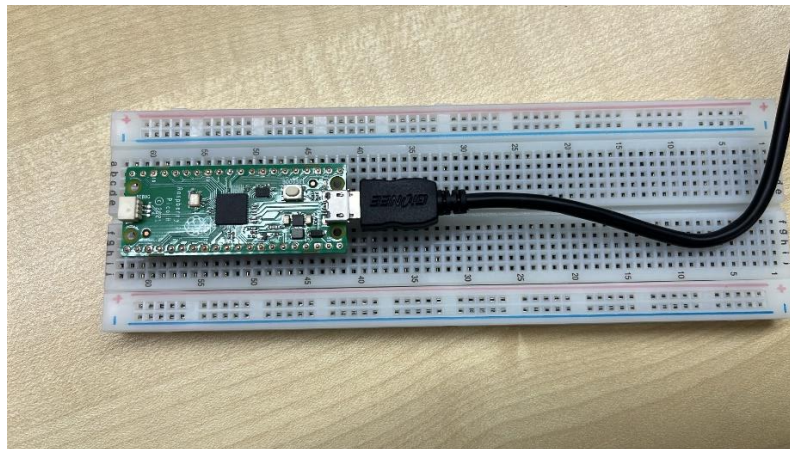
```
1  #include <stdio.h>
2  #include "pico/stdlib.h"
3  #include "hardware/uart.h"
4
5  #define UART_ID uart1
6  #define BAUD_RATE 115200
7
8  #define UART_TX_PIN 4
9  #define UART_RX_PIN 5
10
11 int main()
12 {
13     stdio_init_all();
14
15     uart_init(UART_ID, BAUD_RATE);
16     gpio_set_function(UART_TX_PIN, GPIO_FUNC_UART);
17     gpio_set_function(UART_RX_PIN, GPIO_FUNC_UART);
18     uart_puts(UART_ID, " Hello, UART!\n");
19     while (true) {
20         printf("Hello, Riser Pradip!\n");
21         sleep_ms(1000);
22     }
23 }
```

7. Output:



```
COM4 - PuTTY
Hello, Riser Pradip!
Hello, Riser Pradip!
Hello, Riser Pradip!
Hello, Riser Pradip!
Hello, Riser Pradip!
Hello, Riser Pradip!
Hello, Riser Pradip!
Hello, Riser Pradip!
Hello, Riser Pradip!
Hello, Riser Pradip!
Hello, Riser Pradip!
Hello, Riser Pradip!
Hello, Riser Pradip!
Hello, Riser Pradip!
Hello, Riser Pradip!
Hello, Riser Pradip!
Hello, Riser Pradip!
Hello, Riser Pradip!
Hello, Riser Pradip!
Hello, Riser Pradip!
```

8. Hardware Image:



9. Appendix:

References

- [Official Documentation for Raspberry Pi Pico and Pico W](#)
- [Technical Datasheet for Raspberry Pi Pico](#)
- [GitHub Repository for Raspberry Pi Pico SDK](#)
- [PuTTY Serial Console Documentation](#)